The Nobel Brain Drain

Where Nobel laureates were born and where they were working when they won, 1901 to 2026

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DOI: https://doi.org/10.5281/zenodo.23857487

2026

Abstract

I compare the birthplaces of Nobel laureates with their affiliations at the time of the award, using the official Nobel Prize API for people, prizes and affiliations and Wikidata for coordinates. Of 760 prizes in physics, chemistry, physiology or medicine and economic sciences from 1901 to 2026 with both a present-day birth country and an affiliation, 237 (31%) went to people working outside their birth country, a share that has stayed between 24% and 38% in every decade. The United States is the only large net gainer: 402 prizes were won by people working there and 271 by people born there, a net gain of 131; 34% of its prizes went to people born abroad, and it received 57% of all prizes won outside the laureate's birth country. Germany took 31% of prizes up to 1932 and 7% since 1933; 10% of German-born laureates up to 1932 were working abroad, against 49% since. Japan rose from 5 science laureates in the twentieth century to 22 since 2000, most of them working in Japan. Nine of the ten institutions with most prizes are American. Women won 3.8% of prizes to people before 1960 and 20% in the 2020s so far. Affiliation at the award is not always where the winning work was done, and birth country is not nationality; I report both caveats with the results.

Keywords: Nobel Prize; scientific migration; brain drain; science history; Wikidata; open data; flow maps

1 Introduction

Nobel prizes are a famously narrow measure of science, but a well recorded one, and they have long been used to study where scientific excellence is produced and where it moves (Zuckerman, 1977; Schlagberger et al., 2016). The prizes are claimed by places twice over: by the country where a laureate was born and by the institution where they worked. The gap between the two is a small, visible piece of the history of scientific migration: the European emigration of the 1930s (Medawar and Pyke, 2001), the post-war pull of American universities, and more recent flows from Asia.

The Nobel Prize Outreach organisation publishes an open API that records, for every prize, the laureate's birthplace and their affiliations at the time of the award (Nobel Prize Outreach, 2026). Wikidata adds coordinates for birthplaces and employers (Vrandečić and Krötzsch, 2014). Together they allow a simple, fully reproducible map of every laureate's journey from birthplace to prize, and a country-by-country account of gains and losses. This paper sets out the data and method, then reports the net balance of countries, the role of the United States, Germany before and after 1933, the rise of Japan, the leading institutions, the share of women and the share born abroad over time. It accompanies an interactive map (https://maps.tristanhotham.com/nobel-brain-drain/).

Table 1: From sources to the flow analysis.

Step	Count
Laureates in the Nobel Prize API	1,028
of which organisations (excluded)	28
Prizes awarded to people (records)	1,005
physics, chemistry, medicine, economics	769
with no affiliation in the API	7
with no birthplace in the API	2
in the flow analysis	760
peace and literature	236
with a country of residence	224
Wikidata rows (physics, chemistry, medicine)	2,075
people	666
matched to API records (identifier, prize and year)	666
matched by name, prize and year only	2

2 Data

Nobel Prize API. The API (version 2.1), retrieved in 2026, lists 1,028 laureates, 28 of them organisations. For each person it gives a known name, sex, date and place of birth (city and country as named at the time, and as named now, with coordinates for the present-day city where available), and for each prize the year, category, share, motivation and either the affiliations at the time of the award (name, city, country) or, mostly for Peace and Literature, the country of residence. I treat the API as authoritative for names, prizes, years and affiliations. The downloaded file contains no licence statement; I credit Nobel Prize Outreach and note that the terms of use on nobelprize.org should be checked before reusing the derived laureate list.

Wikidata. A query through QLever (Bast and Buchhold, 2017) returned 2,075 rows for 666 laureates of the physics, chemistry and medicine prizes, with prize, year, place of birth and its coordinates, and employers (P108) with their coordinates or headquarters coordinates. The affiliation property (P1416) came back empty, so employers are used to place affiliations. Wikidata is CC0.

Other sources. Natural Earth (Natural Earth, 2026) countries and populated places; and Wikipedia's "List of Nobel laureates by university affiliation" (revision 1381233532, CC BY-SA 4.0), used only to cross-check institution counts.

3 Methods

3.1 Records and scope

The unit is a prize awarded to a person: 1,005 records for 1,000 people, since 5 people won twice. Organisations are excluded. The flow analysis covers the four prizes for which the API gives affiliations: 769 records, of which 7 have no affiliation (Marie Curie in 1903, Hayek and Myrdal in 1974 and four others) and 2 no birthplace, leaving 760 (99%). Peace and Literature laureates are analysed separately by country of residence. Award decades run from the 1900s (1901 to 1909) to the 2020s (2020 to 2026).

Table 2: How points were placed: birthplaces of all records, workplaces of the flow analysis.

Method	Records
Birthplace	
Wikidata birthplace coordinates	664
Nobel API present-day city point	334
Natural Earth town of the same name	1
Country label point (birthplace vague)	6
No birthplace	2
Workplace (flow analysis)	
Wikidata employer matched to the affiliation	474
Nobel API city point (city centroid)	284
Hand-set point	2

3.2 Matching

Wikidata people were matched to API records by Wikidata identifier, requiring the same prize and year. All 668 physics, chemistry and medicine records matched: 666 by identifier, two by normalised name, prize and year where Wikidata holds a duplicate item. Matching by name alone would have found only 492, because the API's known names often differ from Wikidata labels ("Sir Alexander Fleming"). Two Wikidata rows matched nothing: an item without an English label and the fictional laureate Sheldon Cooper.

3.3 Places and countries

Birth and work countries are the API's present-day countries. Birthplaces are placed at Wikidata's coordinates when they lie within 100 km of the API's city point or inside the API's birth country (664 records); otherwise at the API's city point (334), a Natural Earth town of the same name (1) or the country's label point (6). Where both points exist their median distance is 0.5 km; the largest disagreements are namesake towns in the API (Cluny in France for John Macleod, born in Cluny, Scotland). 99% of placed birthplaces fall inside the API's present-day country on Natural Earth outlines.

The primary affiliation is the first the API lists that has a place; the Howard Hughes Medical Institute and the LIGO/VIRGO Collaboration are listed without a city, so the university listed beside them is used. 83 records list several affiliations and 19 list affiliations in more than one country. The affiliation is placed at a Wikidata employer of the same person whose name matches it (similarity, containment or shared distinctive words) and whose point lies within 60 km of the API's city (474 of 760); otherwise at the API's city point (284) or a hand-set point (2). Table 2 summarises the methods; every row of the released data carries them.

3.4 Measures

A record is *born abroad* when the birth country differs from the country of the primary affiliation. A country's *net gain* is the number of records affiliated there minus the number born there. For the section on Germany, a laureate is German-born when the API names the birth country at the time of birth as Germany or a German state (Prussia, Bavaria, Mecklenburg and others), whatever the present-day country. Institutions are counted for every listed affiliation, as on the Wikipedia list, with campuses of the University of California kept apart and the Rockefeller Institute counted as Rockefeller University.

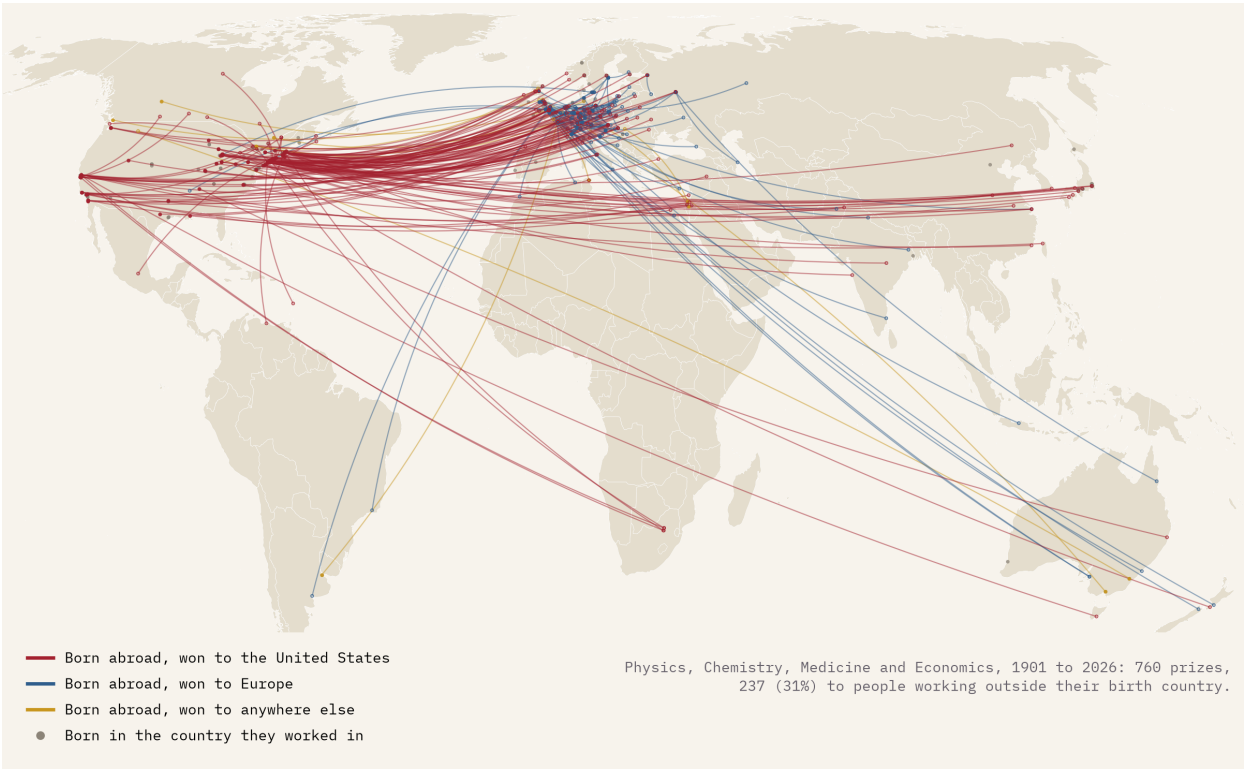


Figure 1: Born-abroad Nobel laureates in physics, chemistry, medicine and economics, from birthplace to affiliation at the award, coloured by where they won. Grey dots: laureates who won in their birth country. Equal Earth projection.

4 Results

4.1 The map

Figure 1 draws every born-abroad record as a curve from birthplace to workplace. The dominant feature is the movement from Europe to the north-eastern United States and California; Figure 2 shows the denser web inside Europe. Overall 31% of records are born abroad. The four subjects are similar: chemistry 34%, physics 32%, economics 31% and medicine 28%. Table 3 lists the largest country pairs.

4.2 Net gain and loss

Only 5 countries have a positive balance (Figure 3, Table 4): the United States (+131), Switzerland (+9), the United Kingdom (+2), Denmark and Sweden (+1 each). The largest losses are Poland (-19), Canada (-9), and Austria, Hungary, the Netherlands and Russia (-8 each). Poland's figure is largely a border effect: 14 of its 19 laureates were born in what was then Germany, and 8 won in Germany. Germany on present-day borders balances exactly (74 born, 74 won there). The United Kingdom both sends and receives: 26 British-born laureates won abroad, 20 of them in the United States, and 28 laureates born elsewhere won in Britain.

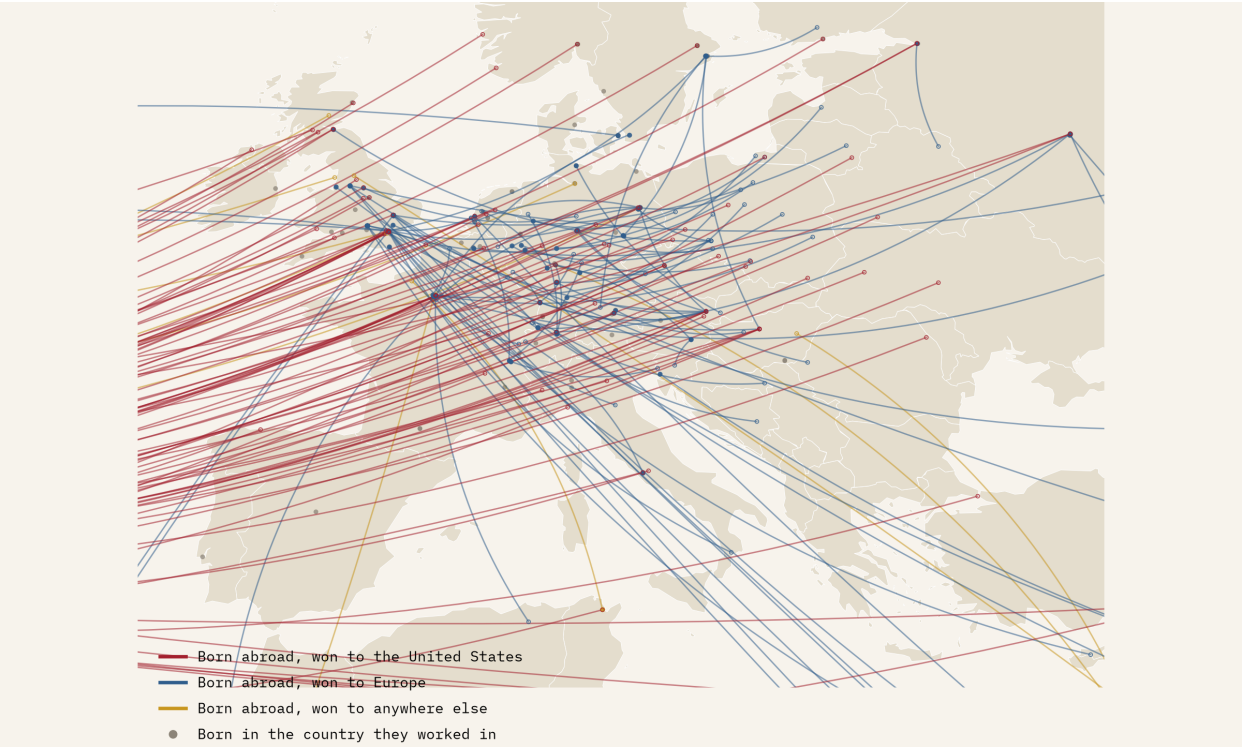


Figure 2: The same flows around Europe.

Table 3: The largest flows from birth country to country of affiliation (present-day borders).

Born in	Won in	Prizes
United Kingdom	United States	20
Canada	United States	15
Germany	United States	12
Poland	Germany	9
France	United States	8
Japan	United States	7
Poland	United States	6
China	United States	6
Russia	United States	5
Italy	United States	5
South Africa	United States	4
Austria	United States	4
Austria	Germany	4
Hungary	United States	4
United Kingdom	Canada	4

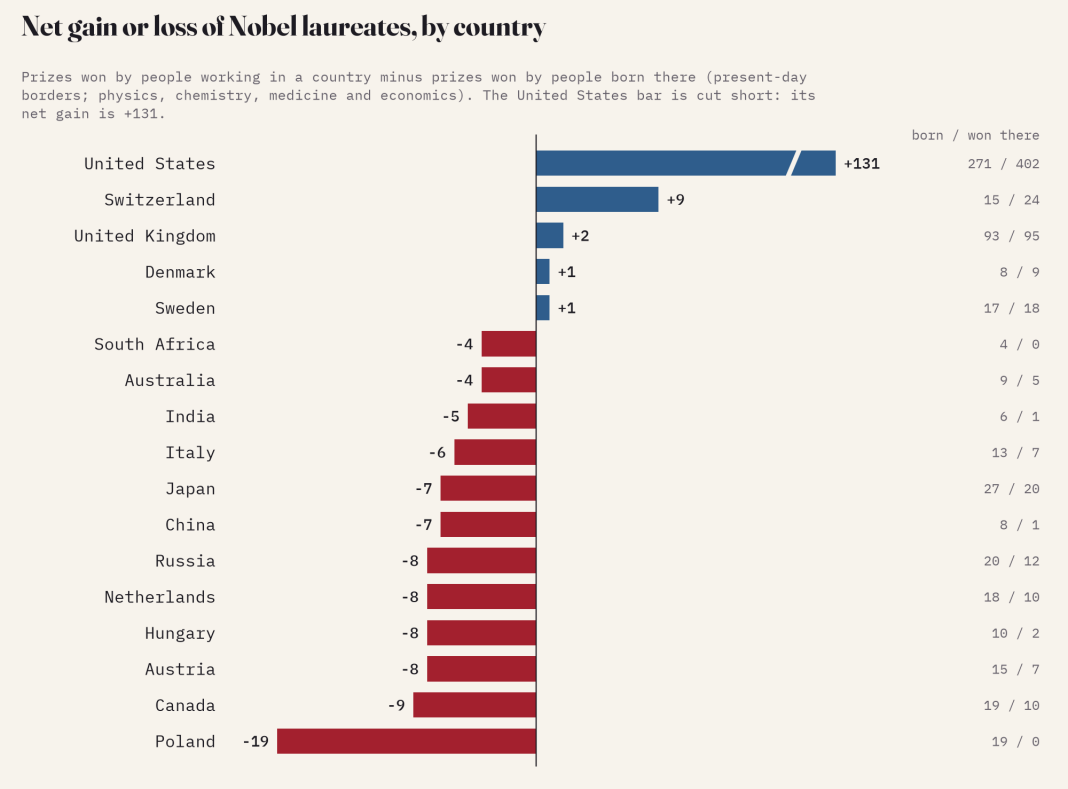


Figure 3: Net gain or loss by country: records affiliated in a country minus records born there. The United States bar is truncated.

Table 4: Countries with most laureates: born there, won there (affiliation at the award), net gain, born there and won abroad, born abroad and won there.

Country	Born	Won	Net	Left	Arrived
United States	271	402	+131	4	135
Switzerland	15	24	+9	3	12
United Kingdom	93	95	+2	26	28
Sweden	17	18	+1	3	4
Denmark	8	9	+1	0	1
Germany	74	74	+0	25	25
Israel	5	5	+0	2	2
Belgium	6	5	-1	2	1
France	42	40	-2	13	11
Norway	8	5	-3	3	0
Czech Republic	4	1	-3	3	0
Australia	9	5	-4	6	2
India	6	1	-5	5	0
Italy	13	7	-6	8	2
Japan	27	20	-7	7	0
China	8	1	-7	7	0
Russia	20	12	-8	11	3
Netherlands	18	10	-8	9	1
Austria	15	7	-8	10	2
Hungary	10	2	-8	8	0
Canada	19	10	-9	15	6
Poland	19	0	-19	19	0

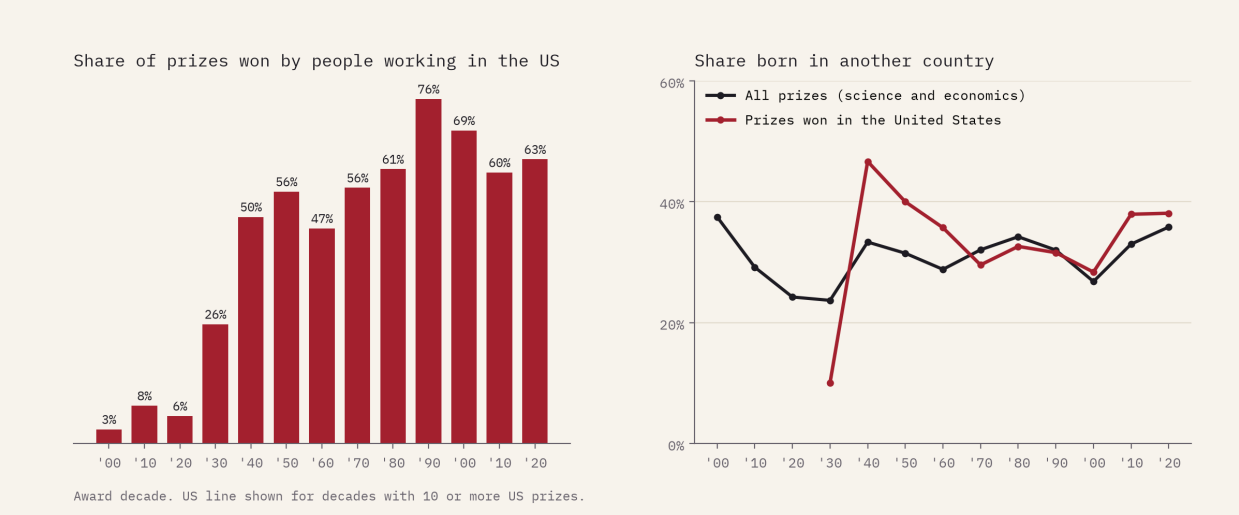


Figure 4: Left: share of each decade's prizes won by people working in the United States. Right: share born in another country, for all prizes and for prizes won in the United States.

4.3 The United States

Germany was the leading country of affiliation in each decade from the 1900s to the 1930s; the United States has led every decade since the 1940s (Figure 4). Its share was 11.8% of prizes before 1940 and 61.7% since 1950. Of the 402 prizes won in the United States, 135 (34%) went to people born in 40 other countries, led by the United Kingdom (20), Canada (15), Germany (12), France (8) and Japan (7). The foreign-born share of American prizes peaked in the 1940s (46.7%), fell to 28.4% in the 2000s and rose to 37.9% in the 2010s and 38.1% in the 2020s so far.

4.4 Germany before and after 1933

Up to 1932, 31 of 99 prizes (31.3%) went to people working in Germany; since 1933, 6.5% (Figure 5). Among the 93 laureates born in the German lands of the day, 10.0% of those who won up to 1932 (30) were working abroad, against 49.2% of those who won from 1933 (63). By birth cohort, the share working abroad rises from 14.8% for those born before 1880 to 37.0% for those born 1880 to 1909 and 61.5% for those born 1910 to 1929. 18 German-born laureates born 1880 to 1929 won abroad (Table 5), 10 in the United States and 4 in Britain. They include well-known refugees from Nazi Germany, but the data do not record when or why anyone moved, and the list also includes people who emigrated as children or after the war.

4.5 Japan

Japan's first laureate was Hideki Yukawa in 1949. Japan had 5 science laureates before 2000 and 22 since (Figure 6); since 2000 it is the third most common birth country, with 8.4% of prizes, against 1.0% before. 7 of 27 Japanese-born laureates won in the United States; the rest won in Japan, at universities and at companies. Yukawa illustrates the limits of affiliation at the award: his meson theory was developed in Japan, but the API lists Columbia University first in 1949.

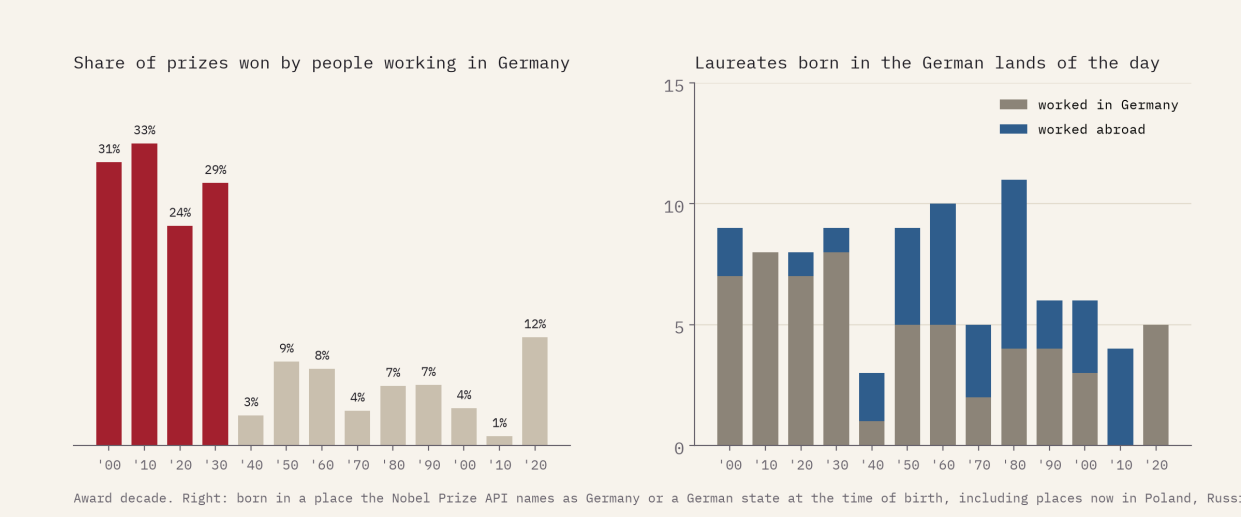


Figure 5: Left: share of each decade's prizes won by people working in Germany. Right: laureates born in the German lands of the day, by where they won.

Table 5: German-born laureates (German lands of the day) born 1880 to 1929 who won while working abroad.

Laureate	Born	Prize	Affiliation at the award
Otto Stern	1888, Sorau	Physics 1943	Carnegie Institute of Technology, United States
Ernst B. Chain	1906, Berlin	Medicine 1945	University of Oxford, United Kingdom
Fritz Lipmann	1899, Koenigsberg	Medicine 1953	Harvard Medical School, United States
Hans Krebs	1900, Hildesheim	Medicine 1953	Sheffield University, United Kingdom
Max Born	1882, Breslau	Physics 1954	Edinburgh University, United Kingdom
Polykarp Kusch	1911, Blankenburg	Physics 1955	Columbia University, United States
Maria Goeppert Mayer	1906, Kattowitz	Physics 1963	University of California, United States
Konrad Bloch	1912, Neisse	Medicine 1964	Harvard University, United States
Alfred Kastler	1902, Guebwiller	Physics 1966	École Normale Supérieure, France
Hans Bethe	1906, Strasbourg	Physics 1967	Cornell University, United States
Max Delbrück	1906, Berlin	Medicine 1969	California Institute of Technology (Caltech), United States
Sir Bernard Katz	1911, Leipzig	Medicine 1970	University College, United Kingdom
Gerhard Herzberg	1904, Hamburg	Chemistry 1971	National Research Council of Canada, Canada
John C. Polanyi	1929, Berlin	Chemistry 1986	University of Toronto, Canada
Jack Steinberger	1921, Bad Kissingen	Physics 1988	CERN, Switzerland
Hans G. Dehmelt	1922, Görlitz	Physics 1989	University of Washington, United States
Herbert Kroemer	1928, Weimar	Physics 2000	University of California, United States
John B. Goodenough	1922, Jena	Chemistry 2019	University of Texas, United States

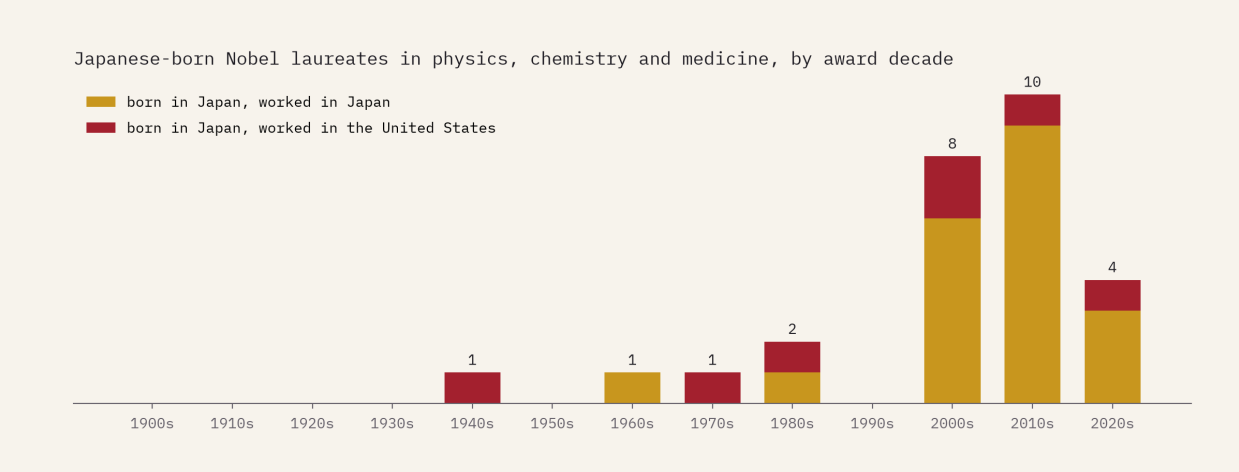


Figure 6: Japanese-born laureates by award decade and where they were working.

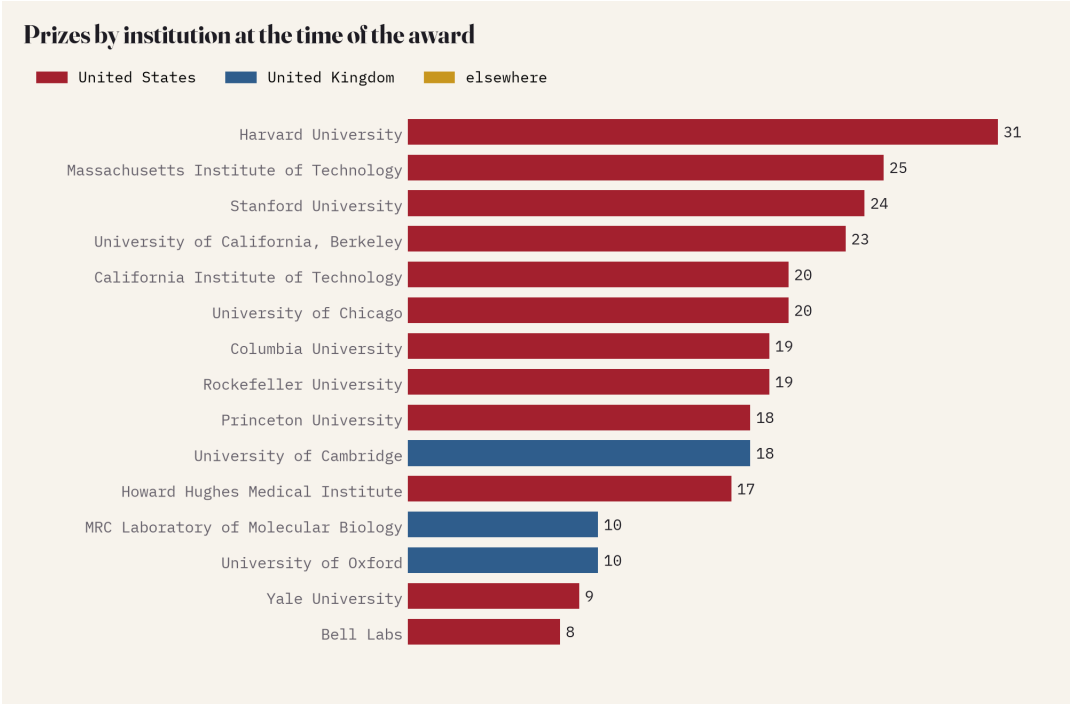


Figure 7: Prizes by institution at the time of the award (every listed affiliation counts).

4.6 Institutions

Counting every listed affiliation, the leading institutions are Harvard University (31), MIT (25), Stanford (24) and the University of California, Berkeley (23); 9 of the top ten are American (Figure 7). Max Planck and Kaiser Wilhelm institutes together appear in 30 records. The counts agree exactly with Wikipedia's list for 15 of its top 20 institutions (Table 6); the rest differ by recent prizes or naming choices.

4.7 Women

Women received 70 of 1,005 prizes to people (7.0%): 12 of 314 before 1960 and 40 of 318 since 2000 (Figure 8). In the four sciences the share rose from 2.2% before 2000 to 7.2% since. Physics has 5 women among 231 laureates, chemistry 8 of 202, medicine 14 of 235, economics 3 of 101, literature

Table 6: Institution counts compared with Wikipedia's list (top 20 there).

Institution (Wikipedia name)	Wikipedia	This study
Harvard University	29	31
Massachusetts Institute of Technology	25	25
Stanford University	24	24
University of California, Berkeley	23	23
California Institute of Technology	20	20
University of Chicago	20	20
Columbia University	19	19
University of Cambridge	18	18
Princeton University	18	18
Howard Hughes Medical Institute	15	17
Rockefeller University	13	19
University of Oxford	10	10
MRC Laboratory of Molecular Biology	10	10
Yale University	9	9
Bell Labs	8	8
Cornell University	8	8
Harvard Medical School	8	8
Case Western Reserve University	7	not matched
Heidelberg University	7	7
University of California, Santa Barbara	7	8

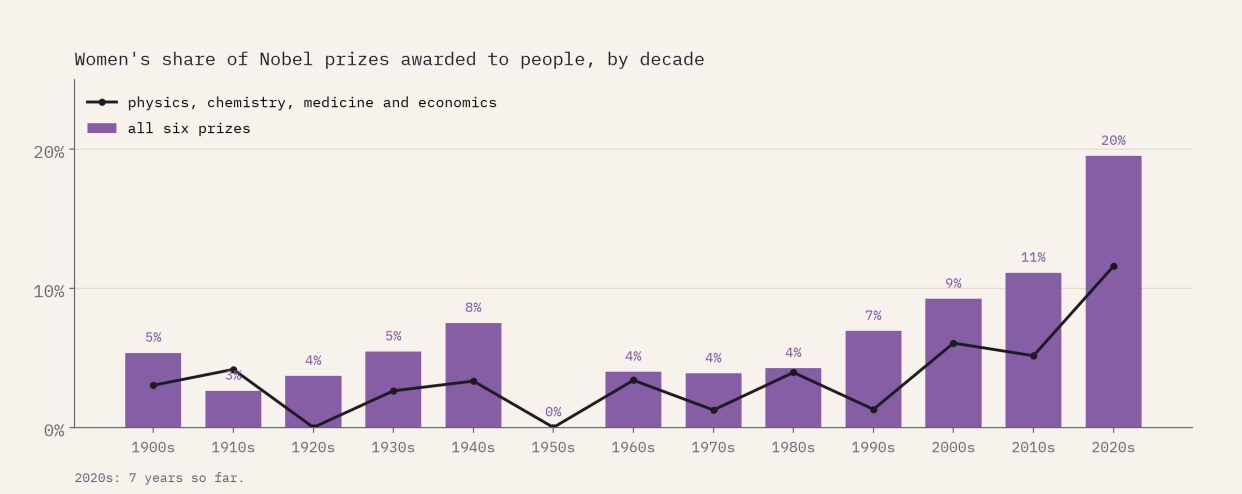


Figure 8: Women's share of prizes to people by decade: all six prizes and the four with affiliations.

19 of 123 and peace 21 of 113. Women in the flow analysis were born abroad about as often as men (27.6% of 29 against 31.3%).

4.8 Born abroad over time

The share born abroad (Table 7) was 37.5% in the 1900s, lowest in the 1930s (23.7%), highest in the 1900s (37.5%) and 35.8% in the 2020s so far. The stability of the share hides a change of direction: early moves were mostly between European countries, later ones mostly to the United States.

Table 7: By award decade: records in the flow analysis, share born abroad, share won in the United States and the foreign-born share of those; prizes to people of all six kinds and the share won by women.

Decade	Flow	Born abroad	In US	US born abroad	All prizes	Women
1900s	32	37.5%	3.1%		56	5.4%
1910s	24	29.2%	8.3%		38	2.6%
1920s	33	24.2%	6.1%		54	3.7%
1930s	38	23.7%	26.3%	10.0%	55	5.5%
1940s	30	33.3%	50.0%	46.7%	40	7.5%
1950s	54	31.5%	55.6%	40.0%	71	0.0%
1960s	59	28.8%	47.5%	35.7%	75	4.0%
1970s	78	32.0%	56.4%	29.6%	103	3.9%
1980s	76	34.2%	60.5%	32.6%	94	4.3%
1990s	75	32.0%	76.0%	31.6%	101	6.9%
2000s	97	26.8%	69.1%	28.4%	119	9.2%
2010s	97	33.0%	59.8%	37.9%	117	11.1%
2020s	67	35.8%	62.7%	38.1%	82	19.5%

4.9 Peace and Literature

The API gives a country of residence at the award for 224 of 236 Peace and Literature laureates. 18.3% of them lived outside their birth country: 23.3% of writers and 13.0% of peace laureates.

5 Discussion

The results restate familiar history in a single, open dataset. Before the 1930s the Nobel map was European and largely German, and movement was mostly between European countries. After 1933 the centre moved to the United States, which since 1950 has hosted most prizes and drawn a third of its laureates from abroad. The German loss appears in the prize record only years later, because prizes typically reward work done long before. Japan stands out as a large late riser that has kept most of its laureates.

The net balance is sensitive to definitions. On present-day borders Germany balances and Poland loses heavily, but most of Poland's loss is Germans born in Silesia and Pomerania. On the API's first affiliation, a laureate holding posts in two countries counts once, though only 19 records are affected.

6 Limitations

First, affiliation at the award is not the place of the prize-winning work, which may be decades earlier. Second, birth country is not nationality or the country of education; many laureates born abroad moved as children. Third, present-day borders reassign historical births. Fourth, the Nobel prizes are a small and selective sample whose committees have their own geographic and gender biases, so the results describe the prize record, not science as a whole. Fifth, placement points are approximate where they fall back to city or country centres, although this affects maps, not country counts. Finally, the API's licence could not be confirmed from the downloaded file.

7 Conclusion

Two open sources are enough to map where every Nobel laureate was born and where they won, and to show that the prize record has one great importer, the United States, and many exporters. The data, code and interactive map are released for reuse and checking.

Data and code availability

The interactive map is at https://maps.tristanhotham.com/nobel-brain-drain/. Data and code are at https://github.com/Tristan-Hotham/nobel-brain-drain and archived with this paper at https://doi.org/10.5281/zenodo.23857487. Derived data, text and figures are released under CC BY 4.0 and code under the MIT licence; Wikidata is CC0, and reuse of the laureate list should follow the Nobel Prize API terms.

AI statement

I wrote the text, with drafting and editing assistance from Claude (Anthropic), which also helped prepare the analysis code. I reviewed and checked all text and results.

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